

A Development Gain That Did Not Survive Locking in LLM-Based Mutation Test Generation

Abstract—Supervised fine-tuning of a 1.5B-parameter code model for mutation test generation raised Kill@8 from 0.605166 to 0.695572 on the panel used to select its checkpoints. On a locked split not used to select a checkpoint, tune a prompt, or set a threshold, the same checkpoint gave +3.4346 percentage points (450/757 $\rightarrow$ 476/757) with an exact two-sided McNemar $p = 0.0783$ against a threshold of $p < 0.01$ fixed before either arm ran; the untrained base model was retained. Over that same comparison, the fraction of candidates passing against the correct implementation fell by 11.8726 points while parse and execution validity did not regress. We report both as negative and inconclusive evidence, and we are explicit that a non-significant difference is not evidence of no difference. We also report that executing the complete measurement path at full scale on a permitted split reproduced a prior Kill@8 of 0.605166 exactly — evidence of execution fidelity, not of model quality.

Index Terms—mutation testing, LLM test generation, negative results, evaluation integrity

I. INTRODUCTION

Mutation kill rate is an attractive score for LLM-generated tests: objective, executable, independent of coverage heuristics. But Kill@k alone can be misleading when candidate and reference validity are not reported alongside it. A test that fails against every implementation still distinguishes a mutant from its reference, so a kill-rate-only evaluation cannot separate a discriminating test from a broken one.

This short paper reports three findings from one controlled study, chosen because each is defensible on its own evidence.

- 1) A selection-panel gain of 9.04 Kill@8 points corresponded to +3.4346 points on a locked split — and the locked result did not meet a predeclared significance threshold.
- 2) Candidate *reference* validity regressed by 11.8726 points over the same locked comparison, while parse and execution validity did not.
- 3) Running the full measurement path at full scale on a permitted split reproduced a previously recorded value exactly, which is what makes the pipeline’s other numbers interpretable.

II. SETTING

The generator is Qwen2.5-Coder-1.5B-Instruct at a pinned immutable revision. All measurements run under one protocol, frozen and identified by digest before use: eight candidates per target, temperature 0.7, top- p 0.9, seed 42, whole-output candidate parsing, raw-output retention, a 1024 token completion limit, no reranking, no feedback rounds.

TABLE I
KILL@8 BY PANEL. THE TWO BLOCKS ARE NOT COMPARABLE
QUANTITIES: THE DEVELOPMENT PANEL CHOSE THE CHECKPOINTS IT
SCORES.

Panel	Arm	Kill@8	Fns	Wilson 95%
<i>Development — selection-only, $n = 542$</i>				
	base	0.605166	328	[0.5634, 0.6454]
	A@431	0.695572	377	[0.6556, 0.7328]
<i>Locked validation — $n = 757$</i>				
	base	0.594452	450	[0.5591, 0.6289]
	A@431	0.628798	476	[0.5938, 0.6625]
<i>Paired locked difference</i>				
	gained / lost / net			+114 / −88 / +26
	Kill@8 delta			+3.4346 points
	McNemar exact p			0.0783
	requirement (predeclared)			$p < 0.01$ — not met
	decision			retain base model

Two panels matter. The *development* panel ($n = 542$ function targets) selected the checkpoints it scores, so its figures are selection statistics and not generalization estimates. The *locked* split ($n = 757$) was not used to select a checkpoint, tune a prompt, or set a threshold; it determined the predeclared promotion decision and was read once. Before either locked arm ran, a promotion rule was committed to a hash-bound receipt requiring all of: Kill@8 gain $\geq +3.0$ points, net gain ≥ 15 functions, and exact two-sided McNemar $p < 0.01$.

A fourth split was reserved for a single final measurement. It was consumed by an authorized attempt that produced **zero candidates and zero metrics**, and may never be rerun. No number in this paper derives from it.

III. RESULTS

A. Development overstated the locked result

Table I and Fig. 1 give the comparison. The selection panel showed +9.04 points; the locked split showed +3.4346, roughly 2.6 $\times$ smaller. This is what selection bias predicts when a panel chooses the checkpoint it then scores.

The locked point estimate is *positive*. With 202 discordant pairs (114 versus 88), an exact two-sided McNemar test [1] gives $p = 0.0783$ against a required $p < 0.01$, so the predeclared criterion failed and the base model was retained. We stress the reading: the data are insufficient to conclude an effect at the standard set in advance. That is **not** a finding of no effect [2], [3], and the question cannot now be settled on this split, which has been used to decide.

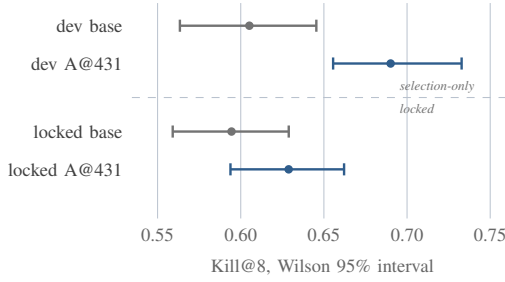


Fig. 1. The locked pair (lower rows) is positive but not significant under the predeclared rule; its intervals overlap. Development and locked rows measure different populations and must not be subtracted.

B. Reference validity regressed

Over the same locked comparison, the fraction of candidates that pass against the *correct* implementation fell from 0.452114 to 0.333388 — a -11.8726 point change — and targets with any reference-valid candidate fell from 675 to 566. Parse validity rose 0.1486 points and execution validity rose 1.3870 points over the same comparison, so this is not general degradation: the tuned model emits *more* usable tests that are *more often* wrong about the reference. The direction held in all four development arms, making it a property of this supervision family rather than of one checkpoint.

A kill-rate-only report would not show this, and it moves opposite to the headline. A tool selected on kill rate alone may be selected against test quality.

C. The pipeline computes what it claims

Executing the complete path — receipt verification, admission, mode-aware scoping, generation, whole-output parsing, safe execution, raw-output retention, progress writing and scoring — at full scale on the permitted panel produced 4,336 candidates over 542 targets with 4,336 retained raw outputs, 4,336 digests, zero missing and zero mismatched, and zero mismatches on an independent recomputation. Its Kill@8 was 0.605166: identical to the value previously recorded for the same model on the same panel under the same evaluation-scope digest. No weights were written and no adapter was loaded.

This is evidence of *execution fidelity*. It is not a model-performance result, not a generalization estimate, not a selection result, and not a substitute for the final measurement that was lost.

That measurement was lost to a narrow defect worth naming, because it is cheap to avoid: the final-run path applied a record-admission rule to *every* record before the scope filter that the established evaluator had always applied first, and so refused records the pipeline was always going to exclude. The component was unreachable until the one-time authorization was spent, and had therefore never been executed against real data. Running the same selector against permitted splits reproduces the rejection immediately, at no cost.

TABLE II
WHAT THIS STUDY DOES AND DOES NOT SUPPORT.

Established

Selection-panel Kill@8 0.605166 $\rightarrow$ 0.695572; locked 0.594452 $\rightarrow$ 0.628798 for the same checkpoint. Paired locked difference $+3.4346$ points, $+114 / - 88$, net $+26$, exact McNemar $p = 0.0783$; predeclared $p < 0.01$ not met; base retained. Reference validity fell 11.8726 points; parse and execution validity did not regress.

A full-scale permitted-split rehearsal reproduced a prior Kill@8 of 0.605166 exactly, with complete raw-output integrity and no weights written.

Inconclusive

Whether fine-tuning improves Kill@8 on unseen data: undetermined at this sample size. Positive point estimate, threshold not met, *not* evidence of no effect.

Whether the validity regression is intrinsic to the objective or specific to this recipe.

Unavailable

No final-test result exists for any arm.

No native repository-level kill rate exists; records of that mode are excluded from every rate reported.

No comparison against any external baseline tool exists; none was bundled or executed in any measurement reported here.

IV. CLAIM STATUS

V. THREATS TO VALIDITY

Construct. Kill@k credits any distinguishing candidate including a universally failing test, which is why reference validity is reported beside it; mutants are an imperfect proxy for real faults [4], [5]. *Internal.* One locked criterion failed through a defect of ours — an adapter-specific digest placed inside a contract required to be identical across arms — recorded as a failure, not reinterpreted; the decision does not rest on it, as the significance criterion failed independently. *External.* One model, one corpus, one supervision family, one seed; no claim that the ratio or trade-off transfers. *Statistical.* A single paired exact test on 202 discordant pairs, reported with Wilson intervals [6], [7], without converting non-significance into a null result.

VI. CONCLUSION

An apparent development gain did not establish reliable improvement on a locked split, and candidate reference validity regressed substantially over the same comparison while parse and execution validity did not. Report a validity metric beside the kill rate; freeze the decision rule before the run; and execute the whole measurement path on permitted data at full scale before anything irreversible depends on it.

DATA AVAILABILITY

An anonymized replication package accompanies this submission. It contains the frozen generation protocol and its digest, the promotion rule as committed before the locked run, the development and locked result receipts, the rehearsal receipt and its sanitised execution receipt, and per-figure data

files each carrying the SHA-256 of its source receipt. Every number in this paper is traceable to one of those receipts.

Raw generated model outputs are retained but excluded from the package because of their volume; their digests are published so any retained output can be verified without redistribution. No material from the consumed split is included, and none exists. The package is hosted on an anonymous-capable archive and contains no repository host, account or link that could identify the authors.

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